

ABSTRACT

Antimicrobial resistance has created an urgent need for the development of alternative therapeutic agents against pathogenic microorganisms. Antimicrobial peptides (AMPs) have gained significant attention due to their broad-spectrum antimicrobial activity, lower susceptibility to resistance development, and ability to target bacterial membranes and virulence-associated proteins. Present study aimed to identify and characterize potential AMPs from publicly available fish transcriptomic datasets using computational approaches and to evaluate their interaction against the sortase A protein, an important virulence factor in Gram-positive bacteria. Candidate peptide sequences were screened based on physicochemical properties including molecular weight, net charge, hydrophobicity, amphipathicity, instability index, and toxicity prediction. The selected peptides exhibited favorable AMP characteristics such as positive net charge, high isoelectric point, amphipathic nature, and non-toxic behavior. Three-dimensional structures of the peptides were predicted using AlphaFold and validated through structural assessment methods. Molecular docking analysis using the HADDOCK server revealed stable peptide–protein complexes with favorable docking scores and interaction energies. Among the selected peptides, Pep80 demonstrated the strongest interaction with Sortase A, exhibiting the lowest HADDOCK score (-84.5 ± 2.0) and a highly stable docking conformation. Visualization and interaction analysis using PyMOL further confirmed effective accommodation of the peptides within the active binding region of sortase A. Overall, the findings suggest that transcriptome-derived fish antimicrobial peptides possess promising therapeutic potential against bacterial virulence targets and may contribute toward the development of novel antimicrobial agents against antibiotic-resistant pathogens.

Keywords: Antimicrobial peptides, Sortase A, AlphaFold, HADDOCK, molecular docking, transcriptomics, fish peptides, antimicrobial resistance.